More on bit-reversal

Khilav Majmudar

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1 Introduction

Carrying on from the Danielson-Lanczos notes, we will discuss some interesting aspects of bit reversal here. We will consider all binary numbers that can be formed with N bits, calling it the set $S(N)$. So,

$$\begin{aligned} S(0) &= \Phi \\ S(1) &= \{0,1\} = \{0,1\} \\ S(2) &= \{00,01,10,11\}_2 = \{0,1,2,3\}_{10} \ldots \end{aligned} \tag{1}$$

where the subscript denotes the base of the set. This will be omitted further below where the distinction is obvious. Now denote by R_N the action of bit-reversal applied to $S(N)$, i.e., $R_N = \texttt{Rev}(S(N))$.

$$\begin{aligned} R_0 &= \Phi \\ R_1 &= \{0,1\} \\ R_2 &= \{00,10,01,11\}_2 = \{0,2,1,3\}_{10} \\ R_3 &= \texttt{Rev}(\{000,001,010,011,100,101,110,111\}_2) \\ &= \{000,100,010,110,001,101,011,111\}_2 = \{0,4,2,6,1,5,3,7\}_{10} \\ R_4 &= \{0,8,4,12,2,10,6,14,1,9,5,13,3,11,7,15\}_{10} \ldots \end{aligned} \tag{2}$$

There are some things to notice here. First, that some numbers reverse to themselves. $R_2(3), R_3(2)$ and $R_4(17)$ are some of them. By $R_N(p)$ we mean the $p\texttt{th}$ element of the set with indexing starting form 0. Second, that every R_N can be formed from R_{N-1} by a simple procedure that has been explained in the earlier document on the Danielson-Lanczos method. To construct the bit-reversed array for $N = p$, take the one for $N = p/2$ and after each element i, insert an index 2^{N-1}. We will rewrite the arrays given above but with the new elements colored red. The elements in black are carried over from the previous N.

$$\begin{aligned} N = 1 &\implies \{0,1\} \\ N = 2 &\implies \{0,2,1,3\} \\ N = 3 &\implies \{0,4,2,6,1,5,3,7\} \\ N = 4 &\implies \{0,8,4,12,2,10,6,14,1,9,5,13,3,11,7,15\} \end{aligned} \tag{3}$$

2 Cycles

Coming back to the first point, let us call the points that reverse to themselves *cycles* [1]. Also, denote the set of cycles of R_N by C_N. Thus,

[1] They are referred to as binary palindromes, and more correctly as fixed points or 1-cycles as they map to themselves after 1 reversing move. We will persist with this terminology until we start dealing with 2-cycles.

$$
\begin{aligned}
C_0 &= \Phi \\
C_1 &= \{0, 1\} \\
C_2 &= \{0, 3\} \\
C_3 &= \{0, 2, 5, 7\} \\
C_4 &= \{0, 6, 9, 15\} \\
C_5 &= \{0, 4, 10, 14, 17, 21, 27, 31\} \\
C_6 &= \{0, 12, 18, 30, 33, 45, 51, 63\} \\
C_7 &= \{0, 8, 20, 28, 34, 42, 54, 62, 65, 73, 85, 93, 99, 107, 109, 127\} \\
C_8 &= \{0, 24, 36, 60, 66, 90, 102, 126, 129, 153, 165, 189, 195, 219, 231, 255\}
\end{aligned} \tag{4}
$$

Cycles can be found by simply finding the symmetric binary numbers for a certain N. We are interested in where these cycles occur in R_N. To see this, we will color the cycles in blue in the sets below. Unfortunately we cannot write the complete R_N neatly in one line unless N is small.

$$
\begin{aligned}
R_1 &= \{0, 1\} \\
R_2 &= \{0, 2, 1, 3\} \\
R_3 &= \{0, 4, 2, 6, 1, 5, 3, 7\} \\
R_4 &= \{0, 8, 4, 12, 2, 10, 6, 14, 1, 9, 5, 13, 3, 11, 7, 15\} \\
R_5 &= \{0, 16, 8, 24, 4, 20, 12, 28, 2, 18, 10, 26, 6, 22, 14, 30, 1, 17, 9, 25, 5, 21, 13, 29, 3, 19, 11, 27, 7, 23, 15, 31\}
\end{aligned} \tag{5}
$$

Obviously, $R_N(2^{N-1}) = 1$ and $R_N(1) = 2^{N-1}$. The two trivial cycles are $0\ldots0$ and $1\ldots1$, which are the first and last elements. One can also spot easily the cycles $100\ldots001$ and $011\ldots110$. The former is two places before the "central" 1 and the latter right after it. See the elements $R_N(2^{N-1} - 2)$ and $R_N(2^{N-1} + 1)$ in the above sets. For example, 14 and 17 in R_5, and 6 and 9 in R_4.

How many cycles are present in R_N? By counting upto $N = 8$, we get the following table.

N	**Cycles**	**N**	**Cycles**
1	2	5	8
2	2	6	8
3	4	7	16
4	4	8	16

Table 1: Number of cycles

Let us denote the number of cycles in R_N, or the number of elements in C_N, as $\mathcal{N}_N$. We may conjecture that if N is even, then $\mathcal{N}_N = 2^{N/2}$, and if N is odd, then $\mathcal{N}_N = 2^{(N+1)/2}$. This can easily be verified by looking at binary numbers in 5 and 6 bits. Take 00100. Finding a cycle under bit-reversal is equivalent to fixing the middle element — the 1 here — and folding one side onto the other. The elements to the left of the middle one can be chosen in $2^{(N-1)/2}$ ways, and the middle element can be either 0 or 1. So, multiplying by two gives $\mathcal{N}_{N_{\texttt{odd}}} = 2^{(N+1)/2}$. For the even N case, consider 001100. Here it is merely a question of folding straight down the middle, so we only need to choose our bits on one side, giving $\mathcal{N}_{N_{\texttt{even}}} = 2^{N/2}$. Combining both odd and even case, we can write the number of cycles for N bits as $\mathcal{N}_N = 2^{\lceil N/2 \rceil}$.

3 Distribution

How are the cycles distributed in R_N? We have already noted the locations of the trivial ones at the two ends of the set and the two close to the "central" 1. Let us write down the number of elements **between** consecutive cycles for the first few N's, represented by **Distances** in Table 2 below.

N	Distances	N	Distances
1	0	5	3,5,3,2,3,5,3
2	2	6	11,5,11,2,11,5,11
3	1,2,1	7	7,11,7,5,7,11,7,2,7,11,7,5,7,11,7
4	5,2,5	8	23,11,23,5,23,11,23,2,23,11,23,5,23,11,23

Table 2: Distance between cycles

We see that the distances between consecutive cycles are symmetric about the middle distance of 2. It may be more useful to write the distances as a mere difference of their position indices, which would increase values in the table above by 1. We represent this distance by **DiffIndex**. The symmetric part to the right of the central distance is denoted by an ellipsis.

N	DiffIndex	N	DiffIndex
1	0	5	4,6,4,3,...
2	3	6	12,6,12,3,...
3	2,3,...	7	8,12,8,6,8,12,8,3,...
4	6,3,...	8	24,12,24,6,24,12,24,3,...

Table 3: Distance between cycle values

We are beginning to see some kind of structure. Apart from the central distance of 3, we see a doubling for each $(2k-1, 2k)$ pair. Now look at the progression from $N = 3$ to $N = 5$ to $N = 7$. We see that we can take a mirror image around the central 3, multiply all distances by 2, reappend the central 3 at the end of that sequence to obtain the next set of differences. Thus, to get the $N = 7$ set, take the $N = 5$ one of $4, 6, 4, 3, \ldots$. Mirroring about 3 gives $4, 6, 4, 3, 4, 6, 4$. Doubling leads to $8, 12, 8, 6, 8, 12, 8$. This gives the first set of distances. To generate the full set, we just need to mirror it after crossing the central distance of 3, giving $8, 12, 8, 6, 8, 12, 8, 3, \ldots$. The same logic holds for going from $N = 4$ to $N = 6$ to $N = 8$.

Now look at the value of the first nontrivial cycle for a $(2k-1, 2k)$ pair. You will see that there is a tripling. Why? What is the smallest cycle we can form after $0 \ldots 0$? Writing down all possible cycles as we will show below for a couple of cases, we will see that it is the binary number with all zeroes except a one in the middle when N is odd and two ones in the middle when N is even. Their values in base 10 are $2^{(N-1)/2}$ for the former and $2^{N/2}$ for the latter. Taking the difference of these two numbers gives us $3(2^{(N-1)/2})$, which is triple the one for $N = 2k-1$. Take a look at the binary numbers taken as examples in the paragraph after Table 1.

4 Generating cycles

Let us now write down the cycles for $N = 7$ and $N = 8$ in both binary and base 10 in Table 4. Look at 4a first. Note that the central bit in the binary numbers alternates between 0 and 1. For every consecutive $0, 1$ pair, we have the same 3-bit binary number — underlined in green in the tables further below — which is mirrored about the central element to make the cycle. What is the value of this underlined binary in base 10? It is shown in blue in the tables below. Now look at equation 5! These are exactly the values in our bit-reversed arrays R_N! For odd N, we reproduce the array $R_{(N-1)/2}$ and for even N we get $R_{N/2}$. There is no central bit for $N = 8$ so the mirror is simply in the middle of the binary sequence. Table 5 shows similar data for $N = 9$ and $N = 10$.

C(10)	C(2)	C(10)	C(2)
0	0000000 (0)	65	1000001 (1)
8	0001000 (0)	73	1001001 (1)
20	0010100 (4)	85	1010101 (5)
28	0011100 (4)	93	1011101 (5)
34	0100010 (2)	99	1100011 (3)
42	0101010 (2)	107	1101011 (3)
54	0110110 (6)	119	1110111 (7)
62	0111110 (6)	127	1111111 (7)

(a) $N = 7$

C(10)	C(2)	C(10)	C(2)
0	00000000 (0)	129	10000001 (1)
24	00011000 (8)	153	10011001 (9)
36	00100100 (4)	165	10100101 (5)
60	00111100 (12)	189	10111101 (13)
66	01000010 (2)	195	11000011 (3)
90	01011010 (10)	219	11011011 (11)
102	01100110 (6)	231	11100111 (7)
126	01111110 (14)	255	11111111 (15)

(b) $N = 8$

Table 4: The central bit in 4a is colored brown, the bits to the right of it are underlined in green and the decimal value of that underlined binary is written in blue in parentheses. There is no central bit in 4b.

C(10)	C(2)	C(10)	C(2)
0	000000000 (0)	257	100000001 (1)
16	000010000 (0)	273	100010001 (1)
40	000101000 (8)	297	100101001 (9)
56	000111000 (8)	313	100111001 (9)
68	001000100 (4)	325	101000101 (5)
84	001010100 (4)	341	101010101 (5)
108	001101100 (12)	365	101101101 (13)
124	001111100 (12)	381	101111101 (13)
130	010000010 (2)	387	110000011 (3)
146	010010010 (2)	403	110010011 (3)
170	010101010 (10)	427	110101011 (11)
186	010111010 (10)	443	110111011 (11)
198	011000110 (6)	455	111000111 (7)
214	011010110 (6)	471	111010111 (7)
238	011101110 (14)	495	111101111 (15)
254	011111110 (14)	511	111111111 (15)

(a) $N = 9$

C(10)	C(2)	C(10)	C(2)
0	0000000000 (0)	513	1000000001 (1)
48	0000110000 (16)	561	1000110001 (17)
72	0001001000 (8)	585	1001001001 (9)
120	0001111000 (24)	633	1001111001 (25)
132	0010000100 (4)	645	1010000101 (5)
180	0010110100 (20)	693	1010110101 (21)
204	0011001100 (12)	717	1011001101 (13)
252	0011111100 (28)	765	1011111101 (29)
258	0100000010 (2)	771	1100000011 (3)
306	0100110010 (18)	819	1100110011 (19)
330	0101001010 (10)	843	1101001011 (11)
378	0101111010 (26)	891	1101111011 (27)
390	0110000110 (6)	903	1110000111 (7)
438	0110110110 (22)	951	1110110111 (23)
462	0111001110 (14)	975	1111001111 (15)
510	0111111110 (30)	1023	1111111111 (31)

(b) $N = 10$

Table 5: Cycles with decimal and binary values

We see that to construct cycles for any even N, we just need the bit-reversed array $R_{N/2}$. A suitable mirroring of the elements of this set leads to C_N. **We don't need to search for all symmetric combinations of bits**. Similarly for odd N. And thus we can write the generator of cycles for N bits as $R_{\lfloor N/2 \rfloor}$. Tables 6 and 7 show similar cycles as above for $N \in \{3, 4, 5, 6\}$.

C(10)	C(2)
0	000 (0)
2	010 (0)
5	101 (1)
7	111 (1)

(a) $N = 3$

C(10)	C(2)
0	0000 (0)
6	0110 (2)
9	1001 (1)
15	1111 (3)

(b) $N = 4$

Table 6: Cycles with decimal and binary values

C(10)	C(2)
0	00000 (0)
4	00100 (0)
10	01010 (2)
14	01110 (2)
17	10001 (1)
21	10101 (1)
27	11011 (3)
31	11111 (3)

(a) $N = 5$

C(10)	C(2)
0	000000 (0)
12	001100 (4)
18	010010 (2)
30	011110 (6)
33	100001 (1)
45	101101 (5)
51	110011 (3)
63	111111 (7)

(b) $N = 6$

Table 7: Cycles with decimal and binary values

5 More on distribution of cycles

5.1 Differences

In this section we will look at our data visually in the form of plots. First we will plot the **DiffIndex** sequence of distances between cycles (see table 3). Figures 1 and 2 are plots of the **DiffIndex** sequence with the x-axis being the position of the distance value. We see our middle distance of 3 with the curve being mirrored about it. Looking at the values and the shapes makes the algorithm of arriving at the next pattern in the even and odd sets becomes clear. This algorithm is explained below table 3. For example, to obtain the curve for $N = 5$, take the points in the $N = 3$ curve, double their difference values, append the value 3 and take a mirror image about that point.

More structures can be seen if we plot these curves in the same graph, as in Figures 3 to 6. Specifically, we become more aware of some *levels*, which are shown as horizontal lines. Figure 3 shows the DiffIndex sequence for consecutive N values, while figure 4 is for even N and figure 5 for odd N. In the even case, the levels are formed from the middle values of 3 and its subsequent doublings, so they are $3, 6, 12, 24, 48, \ldots$. In the odd case we have a more interesting situation. Look at the panel for $N = 3$ in figure 1. We have the presence of the distance of 2 along with 3. Its doublings lead to another set of levels at powers of 2, but they are present only for odd N. Figure 6 is the set of sequences for odd N but the x-axis is truncated to a factor of 8 for ease of viewing. This leads to the curves for the top 3 N being cut.

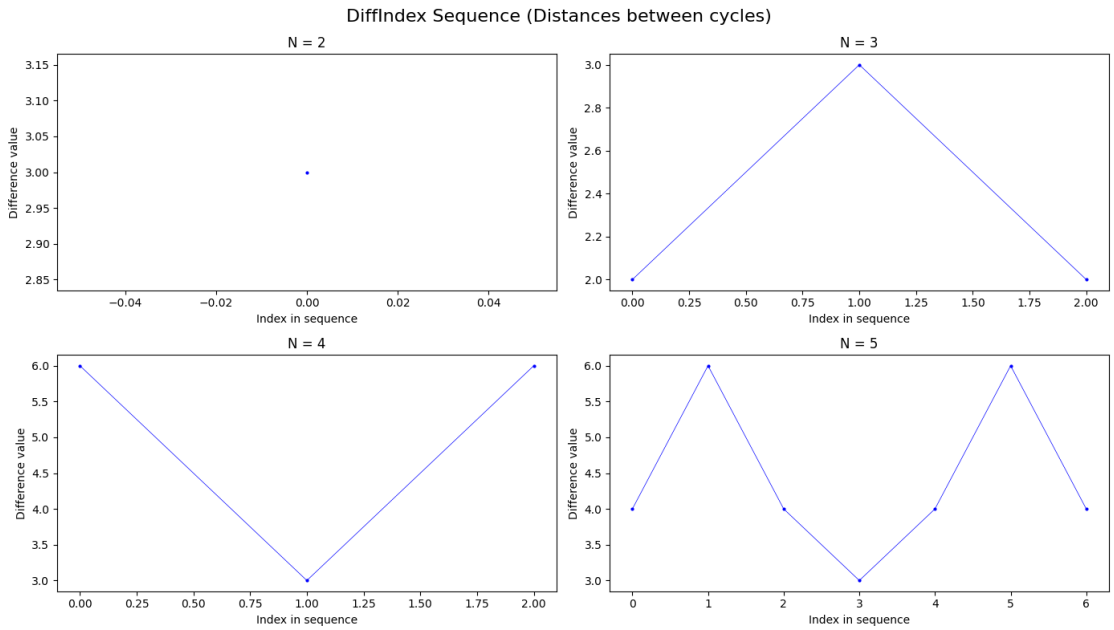

Figure 1

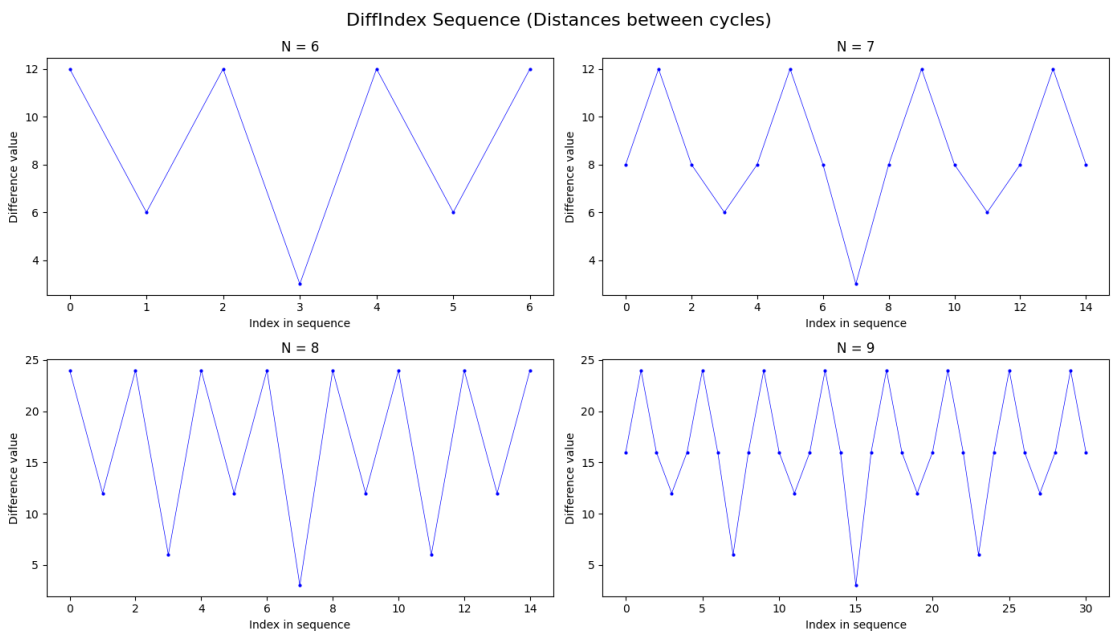

Figure 2

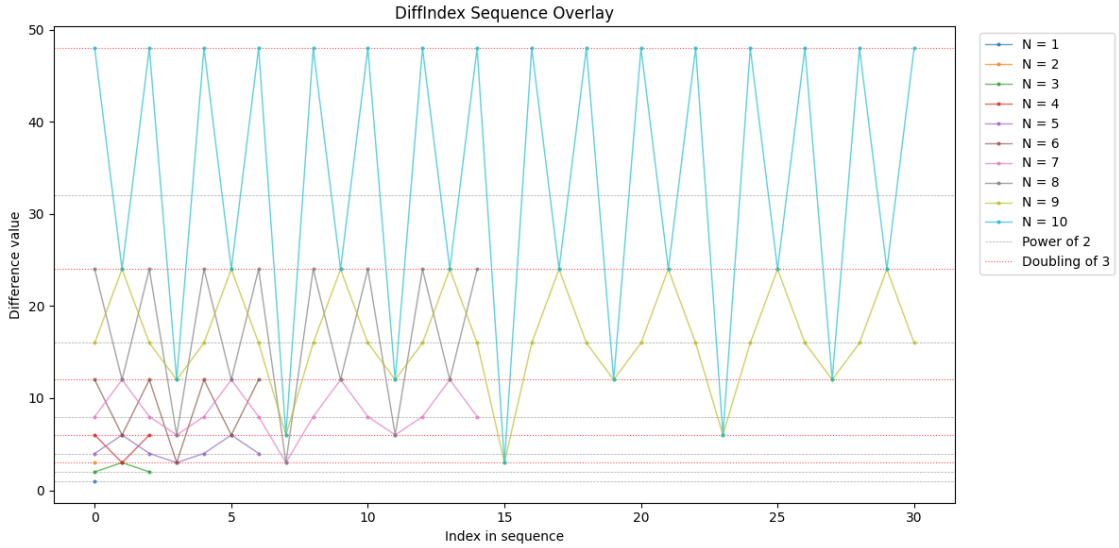

Figure 3

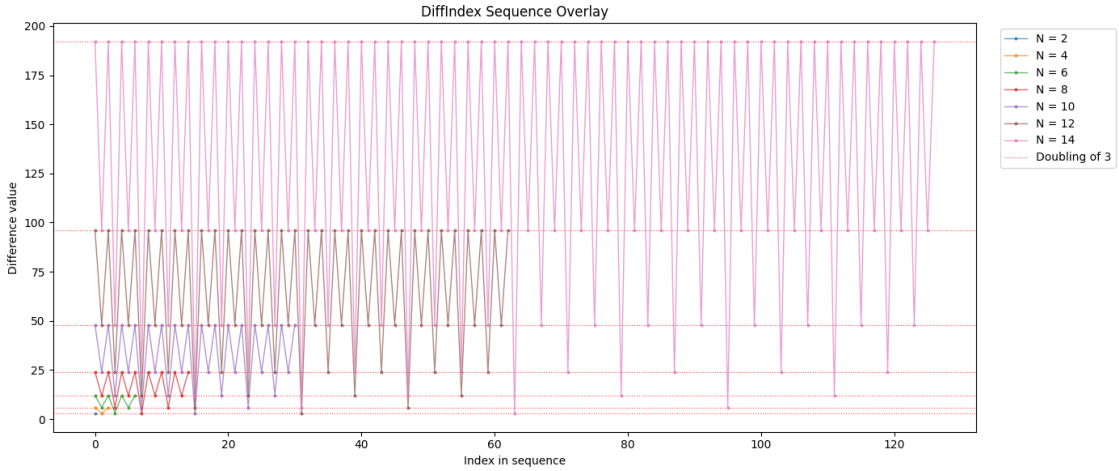

Figure 4

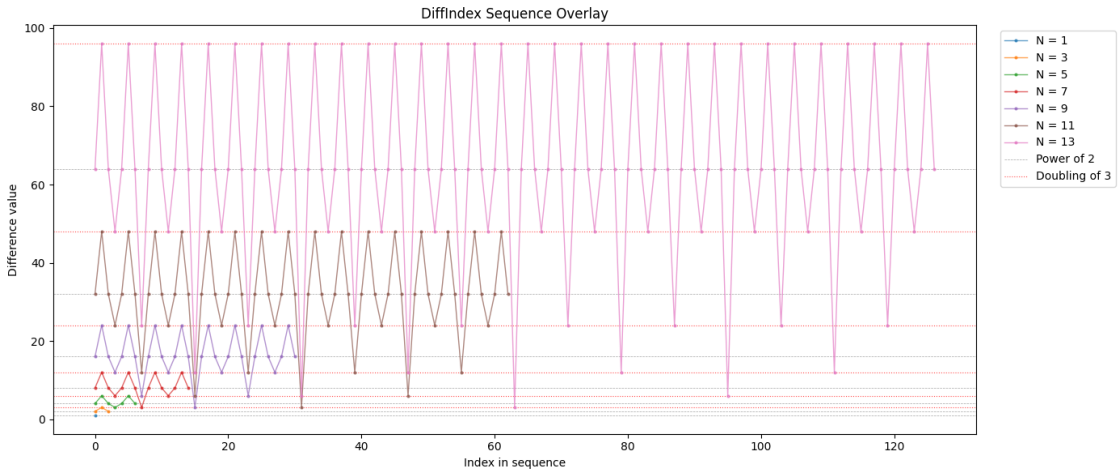

Figure 5

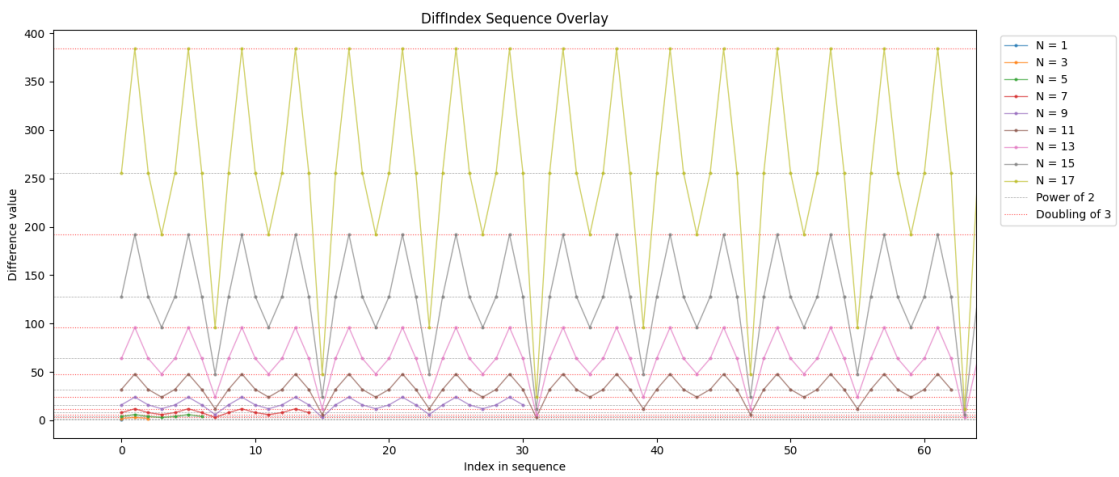

Figure 6

5.2 Distribution of cycles

Here we will simply plot the binary number verses its bit-reversed value, both in decimal. Figure 7 shows this mapping for $N = 7$. Notice that logarithmic scaling to the base 2 has been used for both axes. Some patterns that we will discuss later are visible when we use this scaling instead of the standard base-10 one. Red dots signify our cycles, also called palindromes. The purple dots are the bit-reversed numbers. There is an immediately noticeable obvious symmetry about the $y = x$ line.

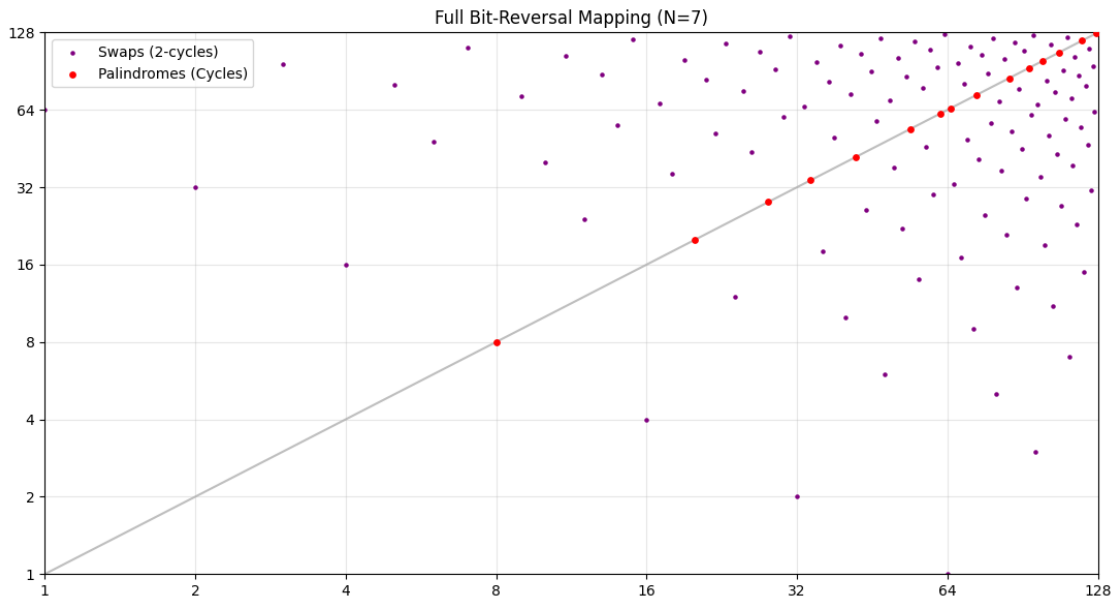

Figure 7

Another feature that can be observed is that with increasing value, the numbers near the top and right boundary seem to follow a curve. This is more evident in figure 8. We see what looks like a kite with curved edges. What is the equation of this edge? It is an easy matter to verify that for hte top sheath it is

$$\begin{aligned} y_k &= 2^{N-k} x_k \\ \text{for} \quad x_k &= 2^k - 1, \quad k \leq N. \end{aligned} \tag{6}$$

Interchange y and x for the equation for the sheath on the right. Thus for the top sheath $x_k = \{1, 3, 7, 15, 31, 63, 127, \dots\}$. These are binary numbers of the form $0\dots01$, $0\dots011$, $0\dots0111$ and so on. This is understandable if we consider their bit-reversals. To make large binary numbers we need as many ones as possible that multiply the larger powers of 2. So we should have y_k's like $10\dots0$, $110\dots0$, $1110\dots0$, etc. The points that make up this sheath are related to another interesting pattern that we will discuss later.

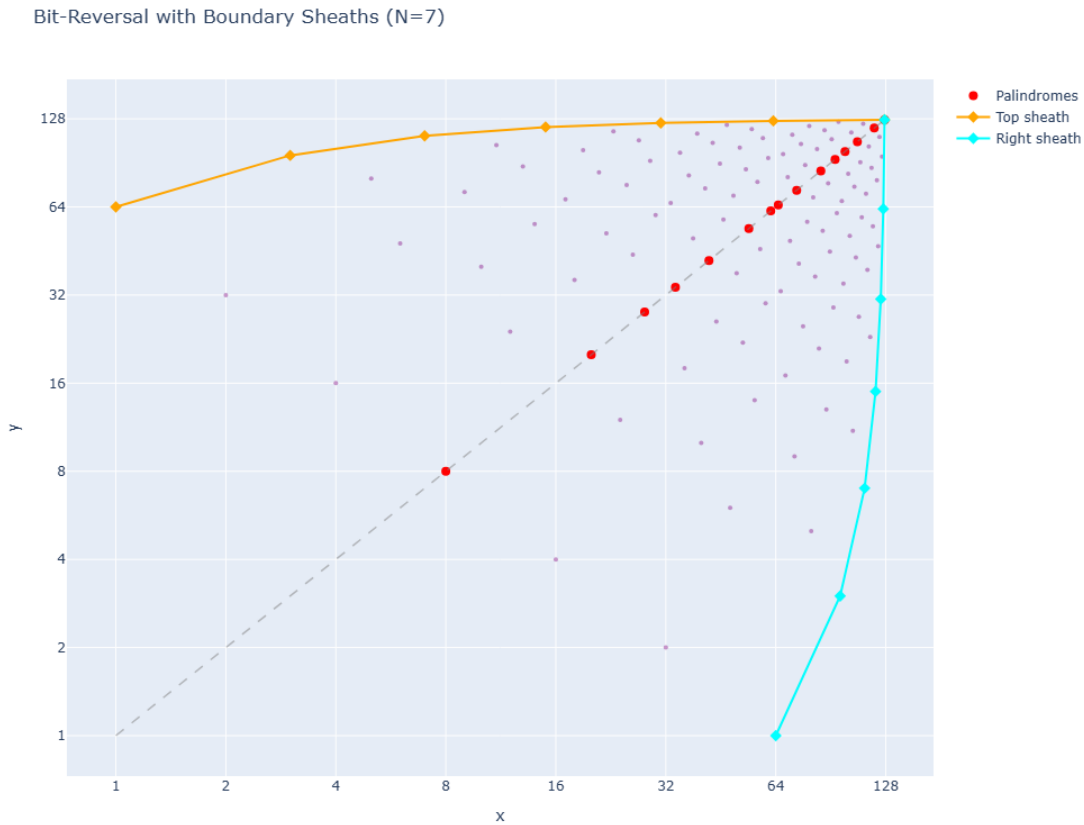
Bit-Reversal with Boundary Sheaths (N=7)
Palindromes
Top sheath
Right sheath
128
64
32
16
8
4
2
1
y
1
2
4
8
16
32
64
128
x

Figure 8